CRS Concepts Explanation

- **Method overloading**
  - In the User class, the constructor is overloaded and allows for one more input variable to be added, the user's birthday. This allows the user to choose whether to also have their birthday data as part of the program if they want.
- **Method overriding** (at least two examples)
  - In the Student class, view_not_full, register1, withdraw, and view_all_course are overridden to define how a student interacts with courses. This lets each subclass provide its own implementation for shared actions.
- **Abstract Class**
  - User is an abstract class. It is used as a blueprint for Student and Admin, allowing shared fields (like username, password) and methods to be defined while forcing subclasses to implement their own specific behaviors.
- **Inheritance**
  - Student and Admin both extend User, inheriting its fields and methods. This reduces code duplicates and lets us treat Student and Admin as User types when needed.
- **Polymorphism**
  - After login, the logged-in user is stored as a User variable, but when methods like menu or others are called, the actual Student or Admin implementation runs.
- **Encapsulation**
  - Fields like username, password, students, etc. are protected or private, and provide constructors and methods to interact with them (like getters and setters). This hides internal details and protects the integrity of the objects.
- **The Concept of ADT** (Abstract Data Types)
  - ArrayList is used to manage collections of students and courses. This helps to add, remove, and search for items without worrying about the underlying implementation.